

Goal: How well does the forward surrogate generalize to Quantum Metal parameters it has never seen before

Probe the in-between regions of SQuADDS by sampling uniformly inside the training cloud and binning by proximity.

1. Sample uniformly

Generate 50,000 uniformly random Quantum Metal parameter sets inside the convex hull of the training data (pure interpolation, no extrapolation).



2. Measure NN distance

For each random point, compute d_{NN} = Euclidean distance to its nearest training sample in the scaled $[0, 1]$ parameter space.



3. Bin and select

Split the d_{NN} range into 10 equal-width bins (close $\rightarrow$ far). Randomly draw 10 points per bin $\rightarrow$ 100 total for EM validation.

Surrogate evaluates all 50,000 samples in seconds.
100 selected across distance bins sent to the EM solver for validation.